

POWER BI PROJECT

INTRODUCTION:

Power BI is a powerful business intelligence tool developed by Microsoft that allows users to visualize, analyze, and share insights from data. In today's data-driven world, organizations rely on clear and interactive reports to make better decisions. Power BI makes it possible to transform raw data into visually appealing dashboards that provide actionable insights.

WHAT IS POWER BI?

Power BI is a data visualization and business analytics tool that connects to various data sources and helps users create interactive reports and dashboards. It combines data preparation, visualization, and sharing in one platform. Supply Chain Performance & Optimization Dashboard This may include:

- **Data Connectivity** – Importing or connecting live data from sources like Excel, SQL, or cloud platforms.
- **Data Transformation** – Cleaning, shaping, and modeling data using Power Query and DAX.
- **Data Visualization** – Creating interactive charts, graphs, maps, and KPI cards.
- **Dashboard Creation** – Building user-friendly dashboards that highlight key business metrics.
- **Data Sharing** – Publishing reports to the Power BI Service for collaboration and decision-making.

WHAT IS POWER QUERY EDITOR?

Power Query Editor is a data transformation tool in Power BI (also available in Excel). It allows users to connect, clean, and shape data before loading it into Power BI for analysis and visualization.

It provides an easy-to-use interface with point-and-click options as well as an underlying M language (Power Query Formula Language) for advanced transformations.

Key Features of Power Query Editor:

- **Data Import** – Connect to multiple sources like Excel, SQL, Web APIs, CSV, etc.
- **Data Cleaning** – Remove duplicates, handle missing values, filter out irrelevant rows, and correct errors.
- **Data Transformation** – Split columns, merge tables, pivot/unpivot, change data types, and create custom columns.
- **Data Shaping** – Reorganize data structures to make them analysis-ready.

- **Applied Steps** – Every transformation is recorded as a step, making the process transparent and reversible.
- **Automation** – Once defined, the same query can be refreshed automatically with updated data.

“SALES DATABASE & OPTIMIZATION DASHBOARD”

INTRODUCTION TO THE SALES DATASET

The Sales Database contains detailed records of product sales transactions from an e-commerce platform (Amazon.in). This dataset is structured to provide a comprehensive view of order processing, shipment details, and sales performance across various product categories and regions in India.

It includes key fields such as **Order ID**, **Order Date**, **Order Status**, **Sales Channel**, **Courier & Fulfilment details**, **Product Information**, and **Shipping Address**. Each row represents an individual customer order, making this database valuable for **sales analysis**, **logistics tracking**, **customer segmentation**, and **performance reporting**.

This dataset was sourced from the Kaggle website, making it publicly available for data analytics, business intelligence, and academic projects.

COLUMN DETAILS – SALES DATASET

1. **Index** – Serial number or row index of each record in the dataset.
2. **Order ID** – Unique identifier for each customer order.
3. **Date** – Order placement date in MM-DD-YY format.
4. **Status** – Current order status (e.g., *Shipped*, *Cancelled*, *Delivered*).
5. **Customer Name** - Name of the customer who made the purchase
6. **Fulfilment** – Indicates whether the order was fulfilled by Amazon or Merchant.
7. **Sales Channel** – Platform through which the order was placed (e.g., *Amazon.in*).
8. **Ship-service-level** – Shipping speed or type chosen for delivery (e.g., *Standard*, *Expedited*).
9. **Style** – Product style code or identifier.
10. **Category** – Type of product category (e.g., *Kurta*, *Top*, *Western Dress*).
11. **Size** – Size of the product (e.g., *S*, *XL*, *3XL*).
12. **ASIN** – Amazon Standard Identification Number — unique product ID.
13. **Courier Status** – Shipping status of the courier (e.g., *Shipped*, *Not Shipped*).
14. **Qty** – Quantity of items ordered.

15. **Amount** – Total order amount in currency.
16. **Ship-city** – Destination city for shipping.
17. **Ship-state** – Destination state for shipping.
18. **Ship-postal-code** – Postal or ZIP code of the shipping address.
19. **B2B** – Indicates whether the order is *Business-to-Business* (True/False).
20. **Ship-postal-code** – Postal or ZIP code of the shipping address.
21. **B2B** – Indicates whether the order is *Business-to-Business* (True/False).

DATA CLEANING & PREPARATION NOTE

The dataset was directly imported into Power BI from Kaggle. Normally, in supply chain analytics, the Power Query Editor is used to clean and preprocess the dataset (e.g., handling missing values, removing duplicates, renaming columns, fixing data types).

However, in this case:

- The dataset contained no missing values.
- There were no duplicates.
- Column data types were correctly assigned (numeric, categorical, etc.).
- Values were already in a clean and consistent format.

DASHBOARD:

